



WORLD DATA AND MISSING MIGRANT, ANALYSIS

SQL PROJECT

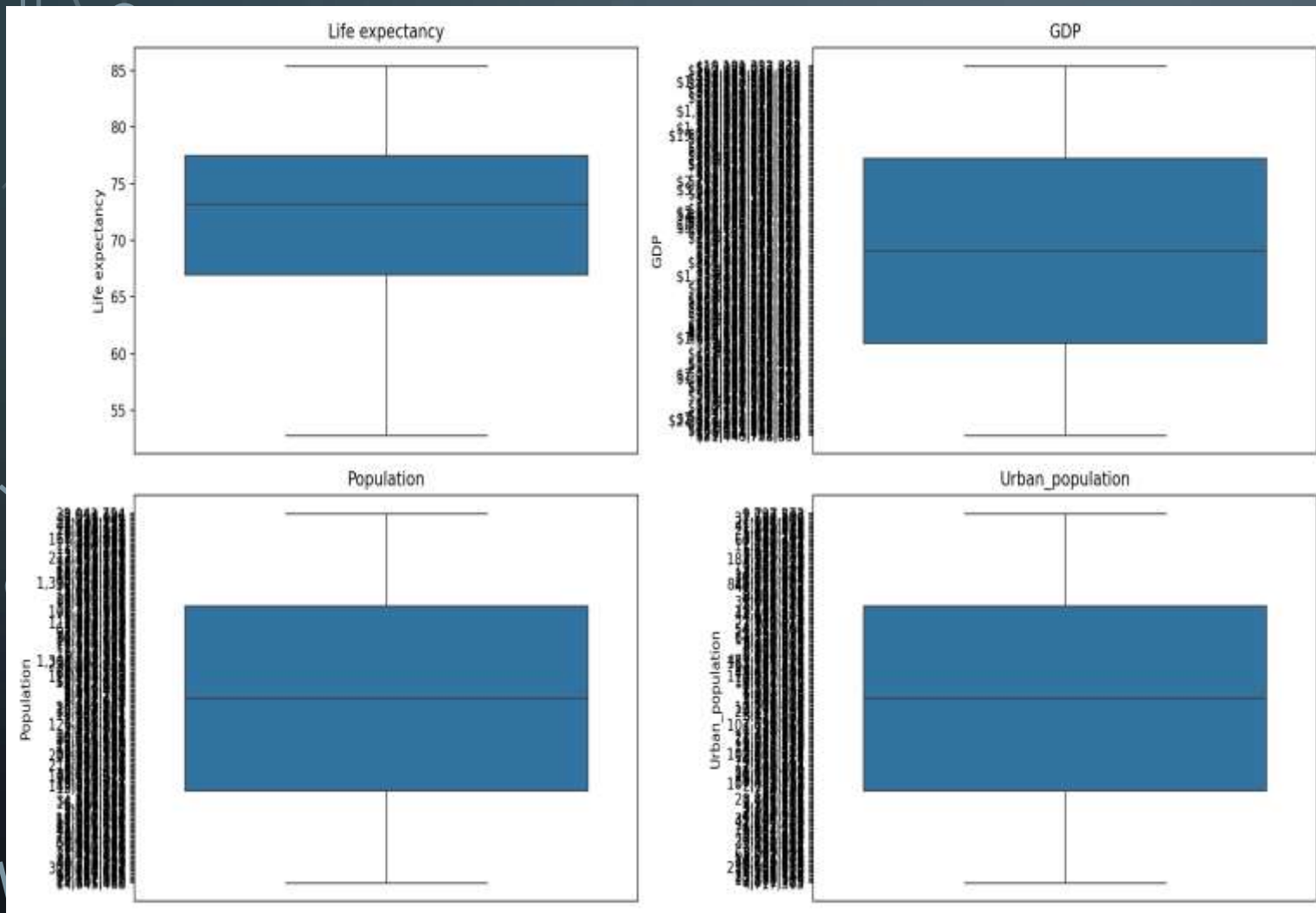
DAVIDE PISTOIA – START2IMPACT

WHAT I DID FOR THIS PROJECT

For this project, I analyzed two datasets: one containing data on countries worldwide and another containing data on global migrations. My goal was to highlight the most important data and attempt to identify correlations.

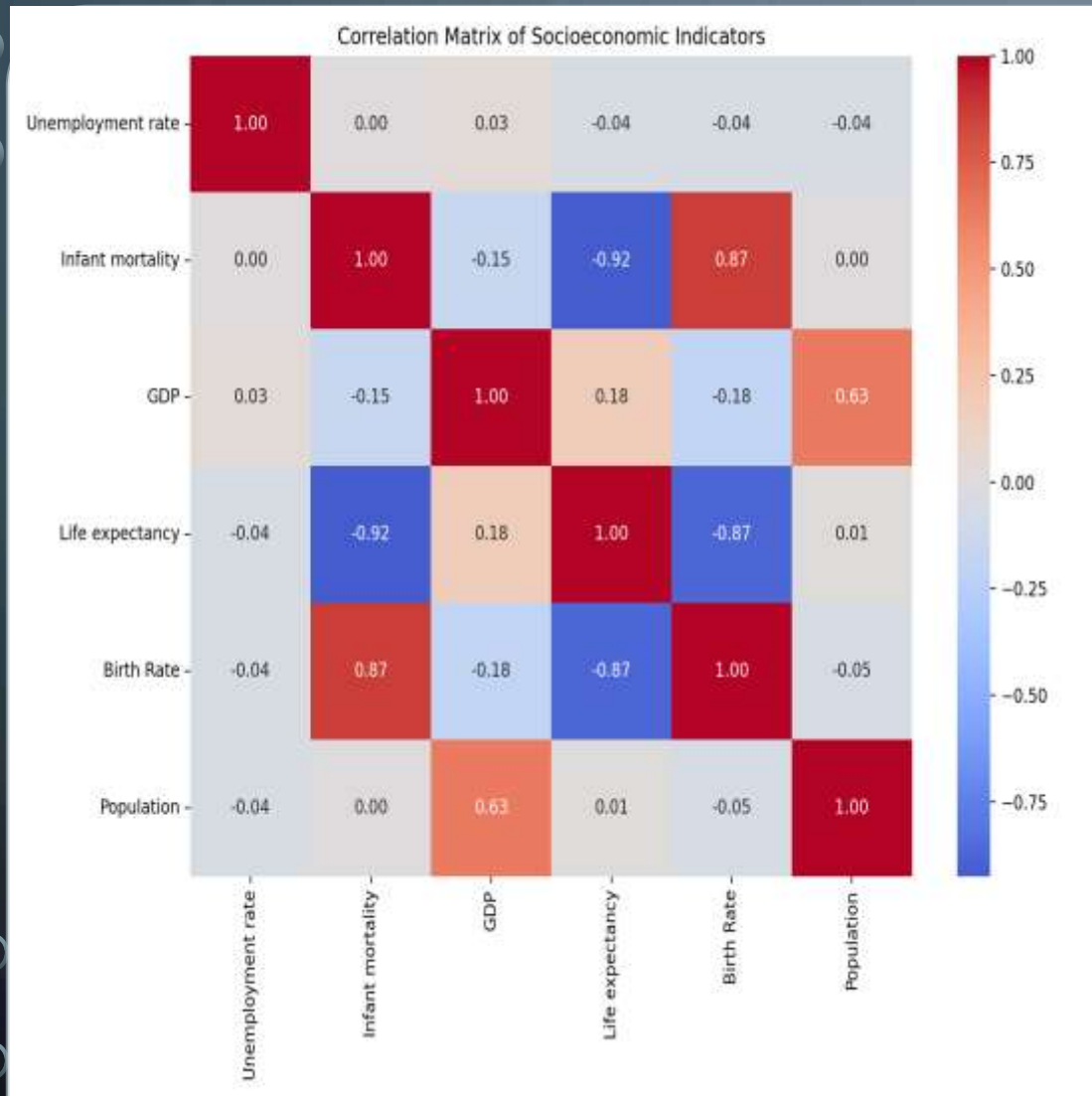
To achieve the goal, I used PostgreSQL to analyze the data. Subsequently, I created charts to better explain the conclusions I reached.

ANALYZE POPULATION DENSITY AND LAND AREA TO STUDY SPATIAL DISTRIBUTION PATTERNS



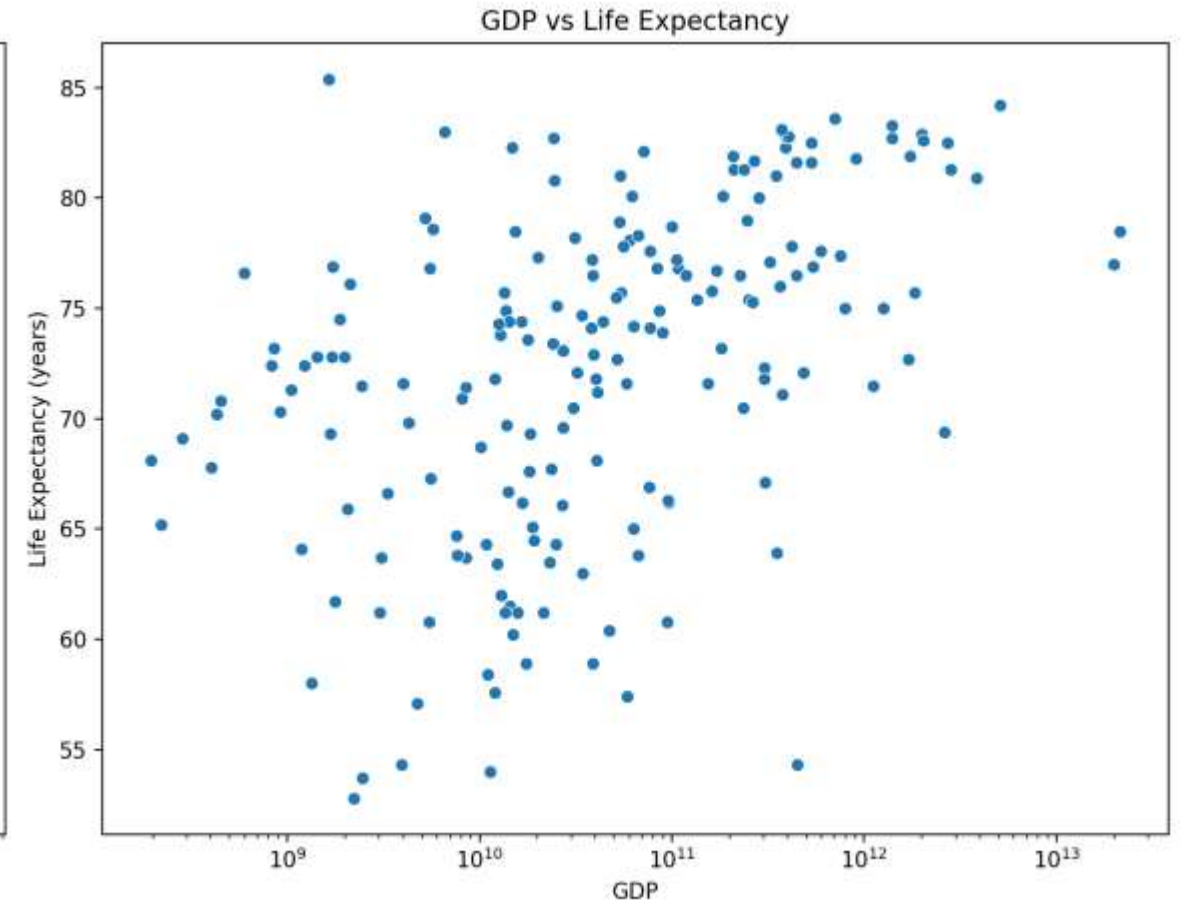
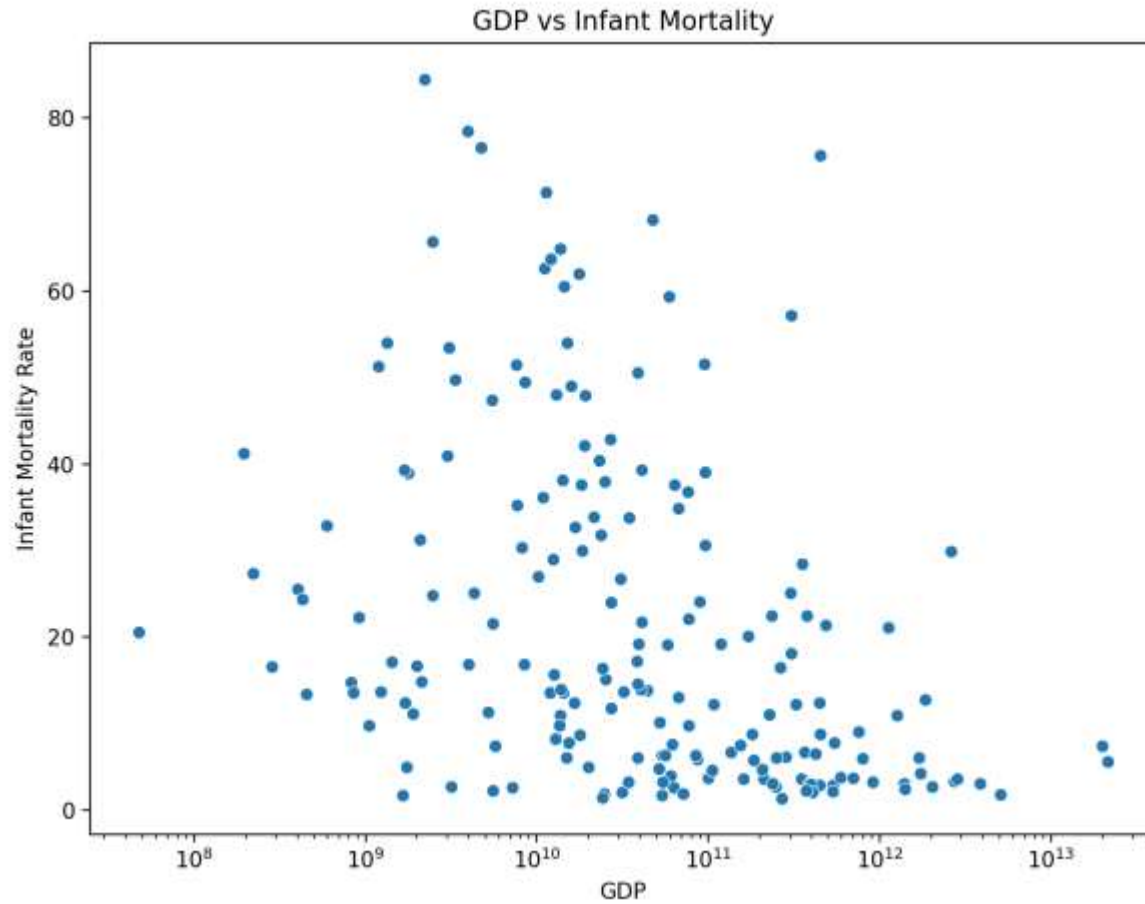
- ⌚ Life Expectancy shows several outliers on both ends, with some countries significantly below the global average
- ⌚ GDP displays extreme outliers on the upper end, indicating a few countries with very high GDP compared to others
- ⌚ Population and Urban Population both show significant right-skewed distributions with multiple outliers

CORRELATIONS BETWEEN ECONOMIC INDICATORS SUCH AS GDP AND VARIOUS SOCIO-ECONOMIC FACTORS. (INFANT MORTALITY, LIFE EXPECTANCY)



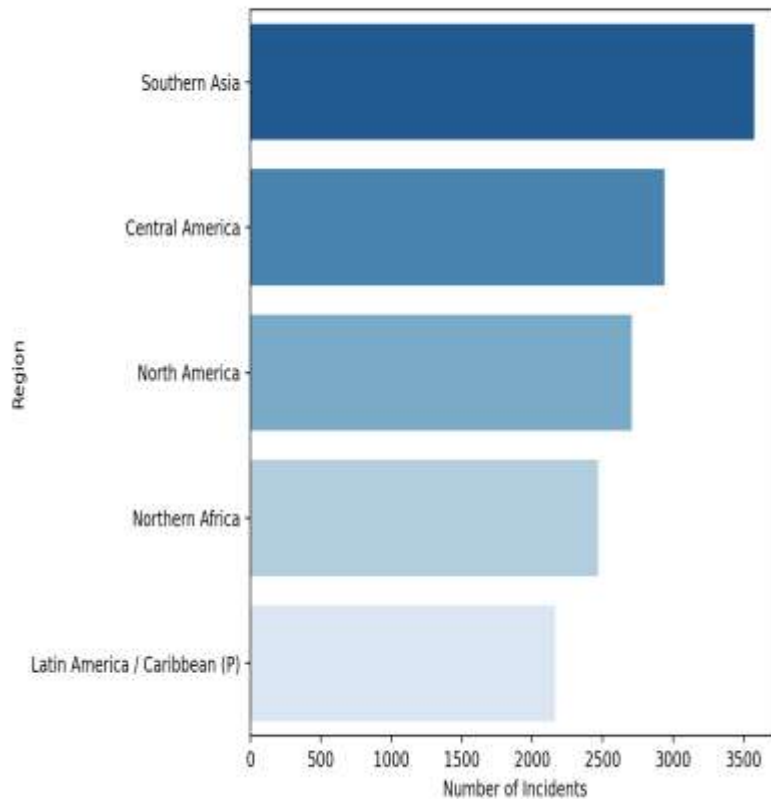
- ⌚ A weak negative correlation between GDP and infant mortality (-0.15)
- ⌚ A weak positive correlation between GDP and life expectancy (0.18)
- ⌚ The scatter plots show that while there are general trends, the relationships aren't strictly linear
- ⌚ Infant mortality steadily decreases as GDP increases, with the most dramatic improvement in the "Very High" GDP quartile (10.58 deaths per 1,000 births) compared to "Low" GDP (28.61)
- ⌚ Life expectancy shows a positive correlation with GDP, increasing from about 69 years in the lowest quartile to 77.5 years in the highest quartile

GDP VS LIFE EXPECTANCY

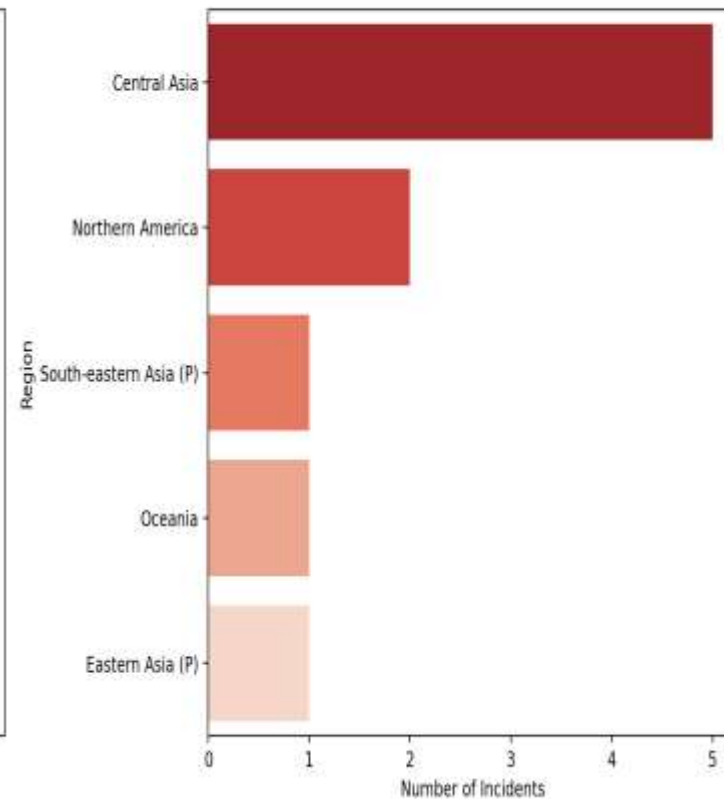


TRENDS AND PATTERNS IN MIGRATION INCIDENTS

Top 5 Most Affected Areas

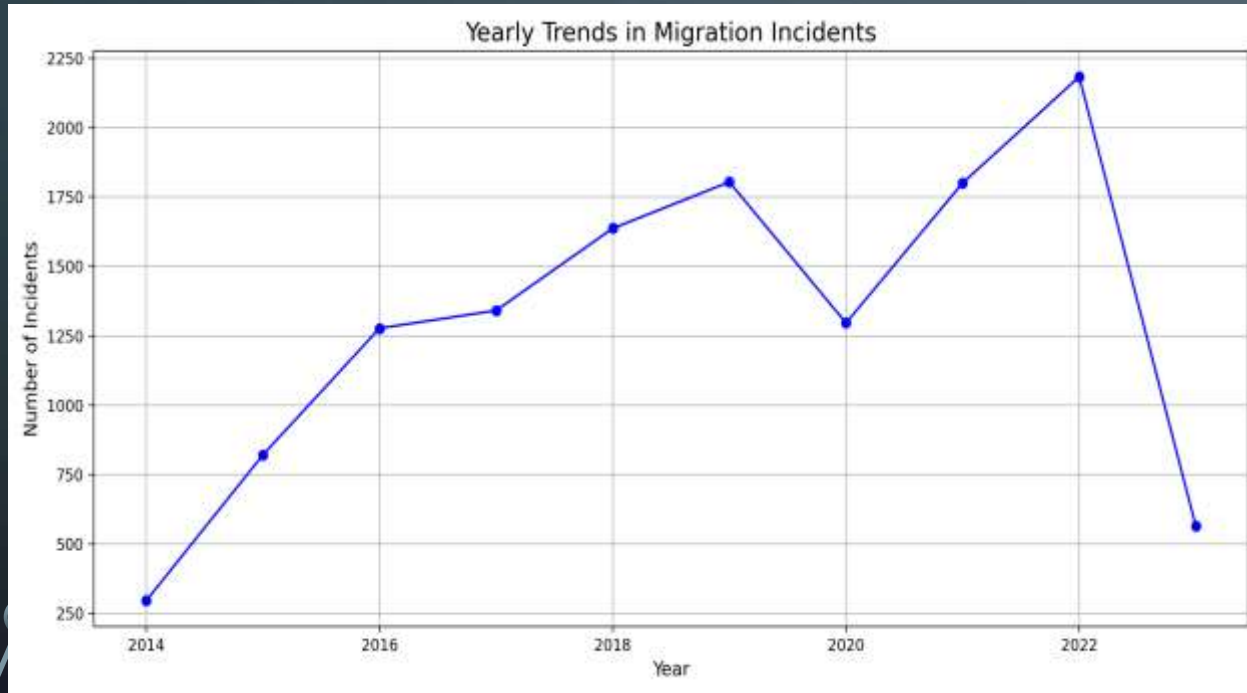


Bottom 5 Least Affected Areas



- These numbers show that Southern Asia has the highest number of migration incidents, followed by Central America and North America. This suggests a significant migration crisis in these regions, with Southern Asia experiencing nearly 3,600 incidents.
- The data shows that regions like Eastern Asia, Oceania, and South-eastern Asia have experienced very few reported incidents, with just 1 incident each. Central Asia and Northern America also show relatively low numbers of incidents.

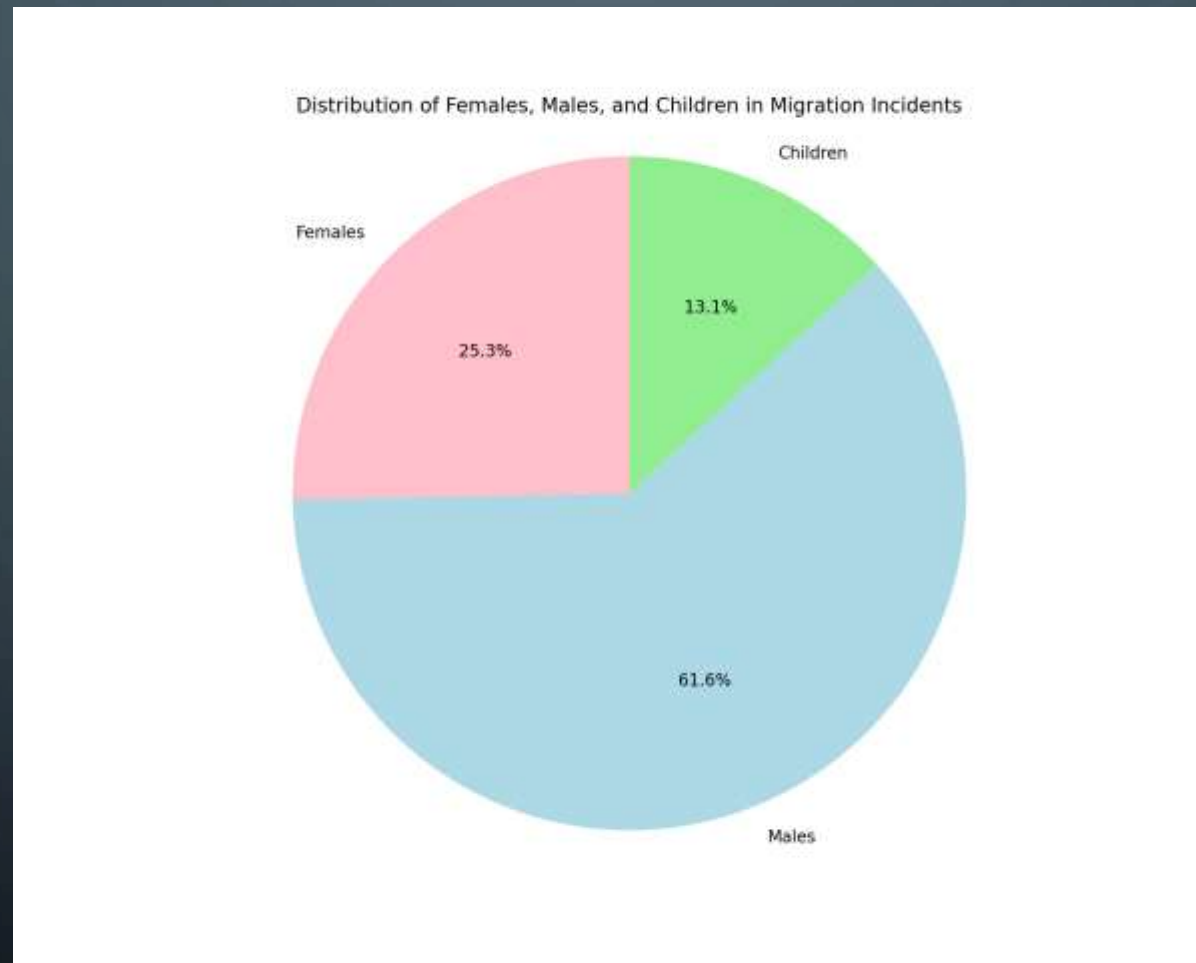
TRENDS OVER TIME, SUCH AS YEARLY INCREASES OR DECREASES IN MIGRATION INCIDENTS



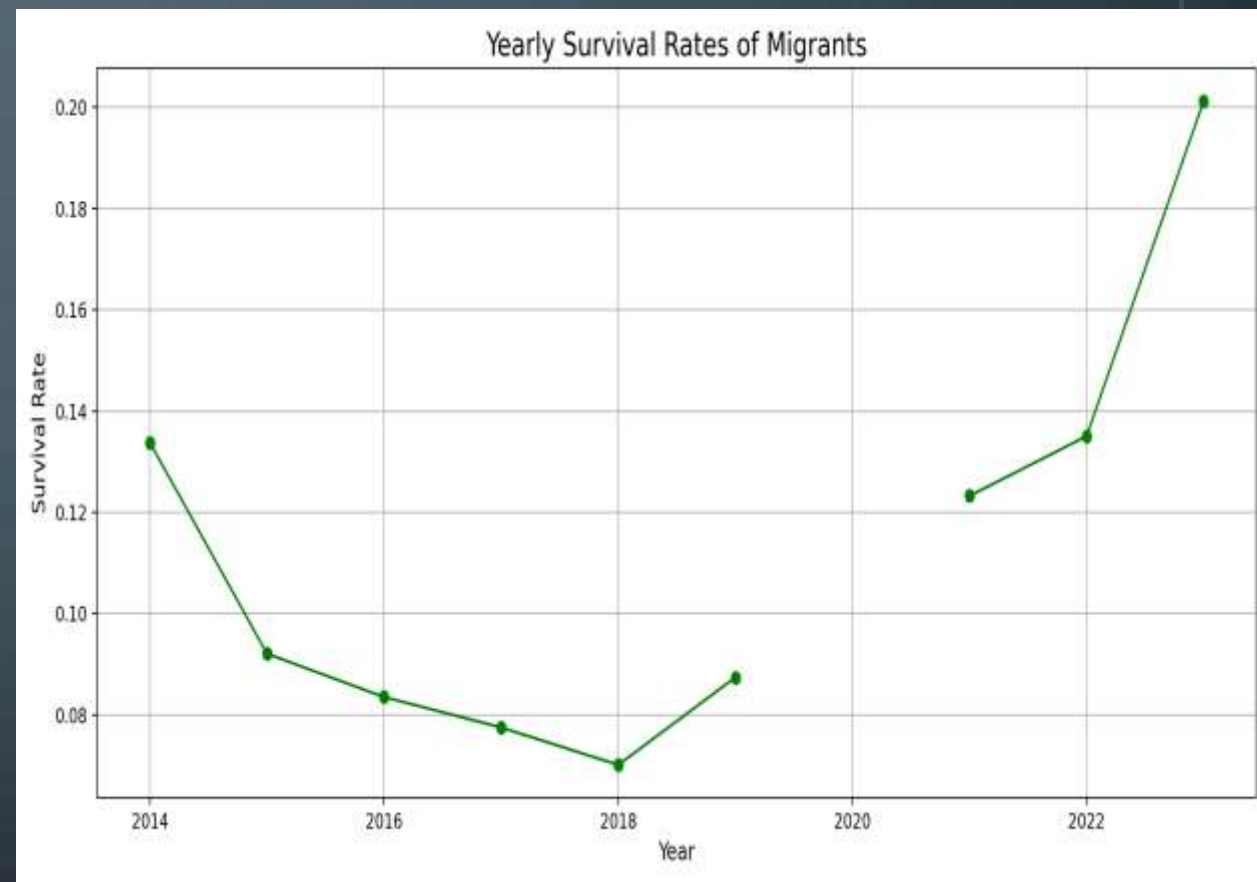
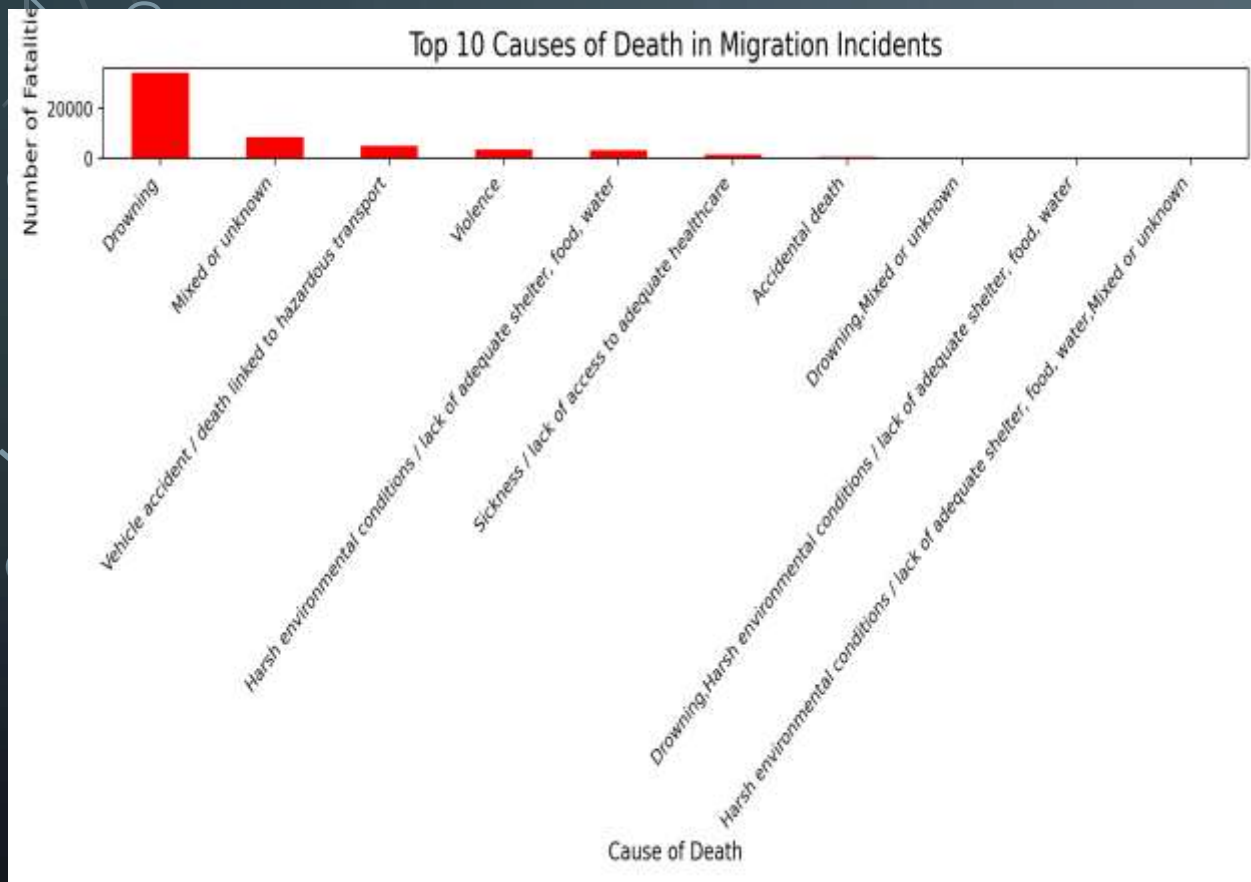
The analysis shows yearly incident counts and percentage changes, identifying 2022 as the peak year with the highest number of incidents and 2014 as the minimum. I will now summarize the trends and their implications. The yearly trends reveal the following:

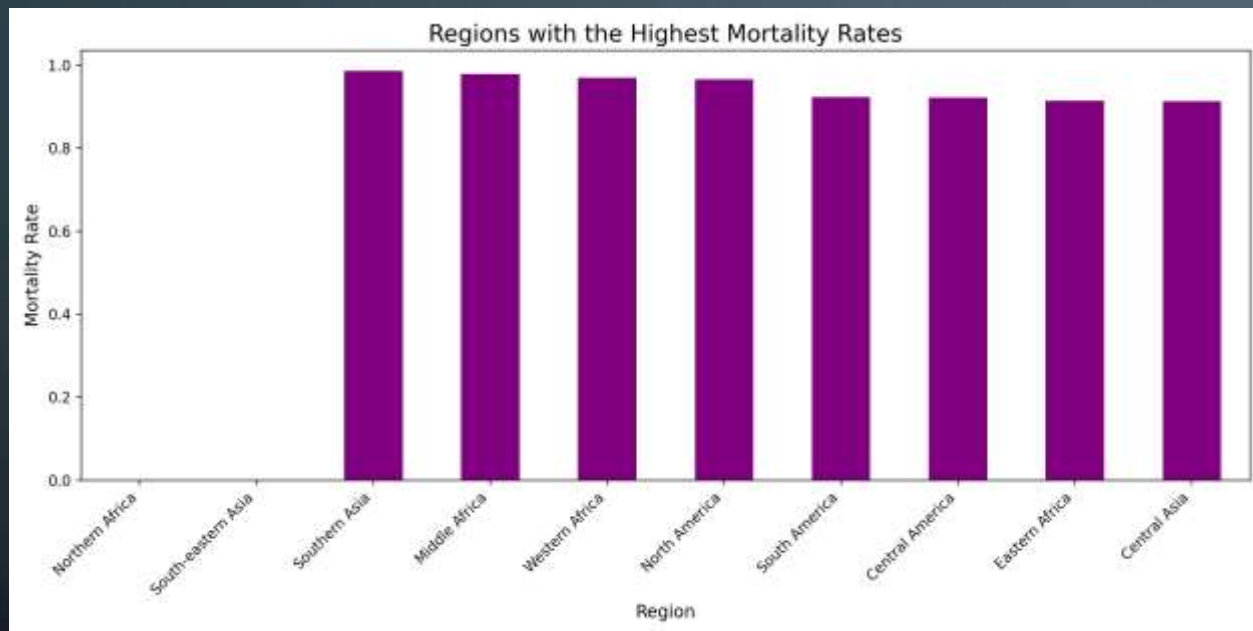
- 🕒 **Peak Year:** Peak year: 2022 with 2183 incidents
- 🕒 **Minimum Year:** Minimum year: 2014 with 296 incidents

TOTAL NUMBER OF FEMALES, MALES, AND CHILDREN INVOLVED IN MIGRATION INCIDENTS



SURVIVAL RATES AND CAUSES OF DEATH TO HIGHLIGHT RISKS AND CHALLENGES MIGRANTS FACE

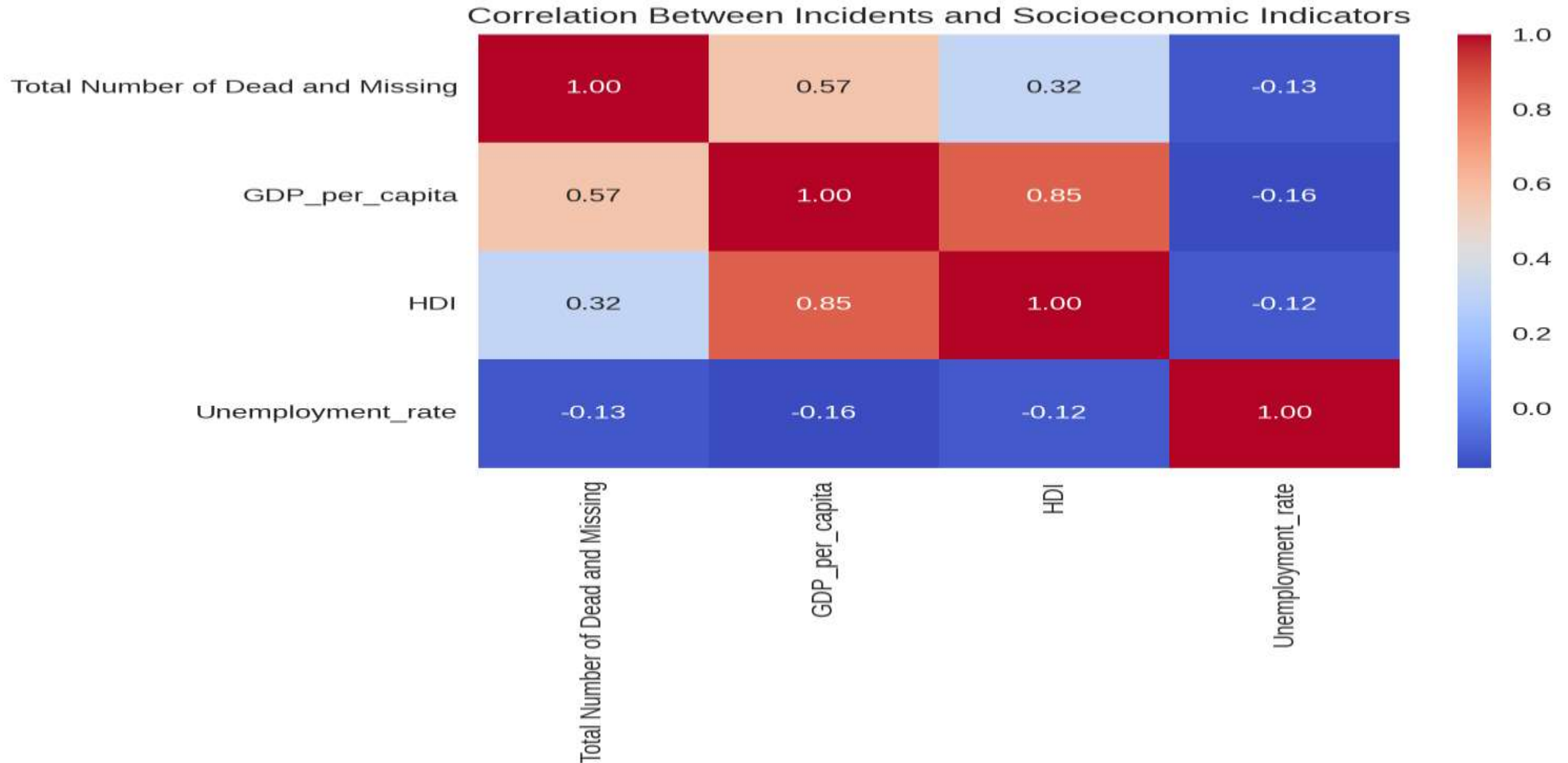




The results highlight key challenges migrants face:

- 1. Survival Rates:** Survival rates have improved slightly over the years, with 2023 showing the highest rate (20%), but they remain alarmingly low overall.
- 2. Causes of Death:** Drowning is the leading cause of death, accounting for the majority of fatalities, followed by hazardous transport and violence.
- 3. Regional Mortality Rates:** Southern Asia and Middle Africa have the highest mortality rates, indicating extreme risks in these regions.

CORRELATIONS ABOUT DATA



THERE ARE SEVERAL SIGNIFICANT RELATIONSHIPS BETWEEN THE SOCIOECONOMIC INDICATORS:

STRONG CORRELATIONS:

- ⌚ LIFE EXPECTANCY AND INFANT MORTALITY HAVE A VERY STRONG NEGATIVE CORRELATION (-0.925), INDICATING THAT COUNTRIES WITH HIGHER LIFE EXPECTANCY TEND TO HAVE LOWER INFANT MORTALITY RATES
- ⌚ BIRTH RATE AND LIFE EXPECTANCY SHOW A STRONG NEGATIVE CORRELATION (-0.875)
- ⌚ BIRTH RATE AND INFANT MORTALITY HAVE A STRONG POSITIVE CORRELATION (0.867)
- ⌚ GDP AND POPULATION SHOW A MODERATE POSITIVE CORRELATION (0.632)

WEAK OR NEGLIGIBLE CORRELATIONS:

- ⌚ UNEMPLOYMENT RATE SHOWS VERY WEAK CORRELATIONS WITH MOST OTHER INDICATORS (ALL CORRELATIONS < 0.05)
- ⌚ GDP HAS WEAK CORRELATIONS WITH MOST INDICATORS EXCEPT POPULATION

THIS ANALYSIS SUGGESTS THAT HEALTH-RELATED INDICATORS (LIFE EXPECTANCY, INFANT MORTALITY, BIRTH RATE) ARE STRONGLY INTERCONNECTED, WHILE ECONOMIC INDICATORS LIKE UNEMPLOYMENT RATE AND GDP SHOW RELATIVELY WEAK RELATIONSHIPS WITH OTHER VARIABLES

THE MEDITERRANEAN REGION HAS RECORDED THE HIGHEST NUMBER OF INCIDENTS, FOLLOWED BY NORTH AMERICA AND NORTHERN AFRICA.

LOOKING AT THE CASUALTIES BY REGION, THE MEDITERRANEAN ROUTE HAS BEEN PARTICULARLY DEADLY WITH OVER 27,000 TOTAL DEAD AND MISSING PERSONS.

THE US-MEXICO BORDER CROSSING IS THE MOST FREQUENTLY DOCUMENTED MIGRATION ROUTE IN THE DATASET, FOLLOWED BY THE SAHARA DESERT CROSSING AND CENTRAL MEDITERRANEAN ROUTE.

THE CORRELATION ANALYSIS REVEALS THE FOLLOWING:

- ⌚ A MODERATE POSITIVE CORRELATION (0.57) BETWEEN GDP PER CAPITA AND THE NUMBER OF INCIDENTS, SUGGESTING THAT WEALTHIER COUNTRIES MAY EXPERIENCE MORE REPORTED INCIDENTS DUE TO BETTER REPORTING MECHANISMS OR PROXIMITY TO MIGRATION ROUTES.
- ⌚ A WEAKER POSITIVE CORRELATION (0.32) BETWEEN HDI AND INCIDENTS, INDICATING A SLIGHT ASSOCIATION WITH HUMAN DEVELOPMENT.
- ⌚ A WEAK NEGATIVE CORRELATION (-0.13) BETWEEN UNEMPLOYMENT RATE AND INCIDENTS, IMPLYING THAT HIGHER UNEMPLOYMENT DOES NOT STRONGLY CORRELATE WITH INCIDENT FREQUENCY.